Definition (Functorial stacking semicategory)
A *functorial stacking semicategory* is a stacking semicategory where the two stacking operations $\otimes$ and $\otimes$ are the two components of a semifunctor

$$\otimes : \mathbf{C} \times \mathbf{C} \to \mathbf{C}.$$

Concretely, and in infix notation, this means that

$$(f \fatsemi h) \otimes (g \fatsemi i) = (f \otimes g) \fatsemi (h \otimes i)$$

for all morphisms f, g, h, and i where respectively f and h, and g and i are composable.